



**The University of Azad Jammu and Kashmir,  
Muzaffarabad**

**Name: Muneeba Arshad**

**Roll No: 2024-SE-06**

**Course Title: Computer Networks**

**Course Code: SE-2203**

**Lab No : 10**

**Submitted To: Engr. Dr. Asma Javaid**

*Department Of Software Engineering*

## ✓ **Inter-VLAN Routing Using Router-on-a-Stick**

*Cisco Packet Tracer – Complete Configuration and Verification*

### **1. Objective**

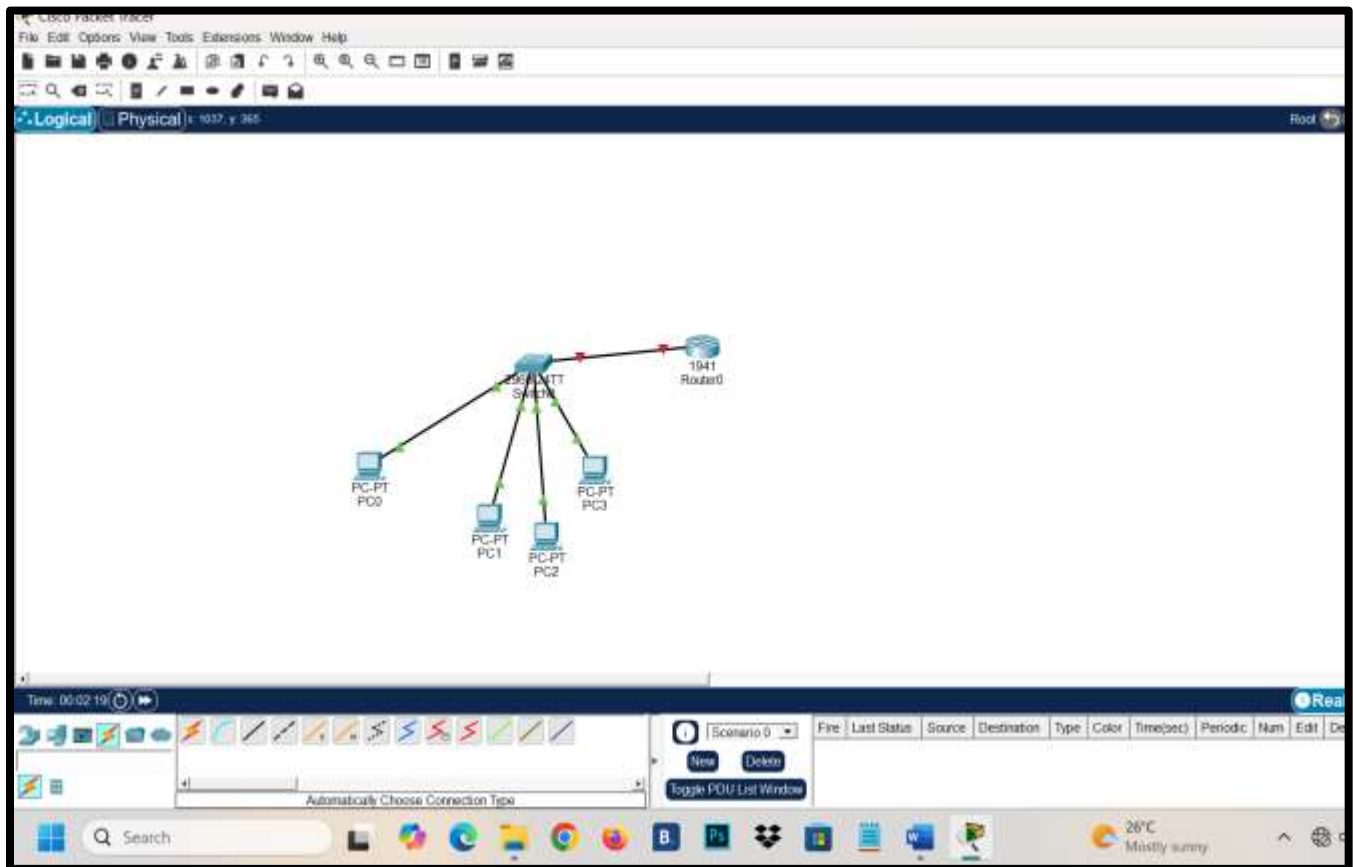
Create VLAN 10 (DEV) and VLAN 20 (HR), assign PCs to their VLANs, configure the switch-router link as a trunk, configure router subinterfaces, assign IP addresses and default gateways, and test communication between the VLANs.

### **2. Create the Topology**

Place 1 router, 1 switch, and 4 PCs in Cisco Packet Tracer. Connect the PCs to the switch and connect the switch to the router as shown in Figure 1.

- PC0 → Switch Fa0/1
- PC1 → Switch Fa0/2
- PC2 → Switch Fa0/3
- PC3 → Switch Fa0/4
- Switch Fa0/24 → Router GigabitEthernet0/0

Brief description: The switch connects the PCs, while the router provides Layer-3 communication between the two VLANs.



### 3. Create VLAN 10 and VLAN 20

Brief description: VLAN 10 separates Development traffic and VLAN 20 separates Human Resources traffic.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name DEV
Switch(config-vlan)#exit
```

### 4. Assign PCs to Their VLANs

Brief description: Fa0/1–Fa0/2 are access ports for DEV; Fa0/3–Fa0/4 are access ports for HR.

```
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
```

```
Switch(config)#interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
```

## 5. Configure the Switch-Router Link as a Trunk

```
interface fa0/24
switchport mode trunk
exit
```

```
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
```

Brief description: The trunk allows VLAN 10 and VLAN 20 traffic to travel over the same physical link to the router.

## 6. Configure Router Sub-interfaces

```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to u

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

Top

```
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

Brief description: Each subinterface is the default gateway for its VLAN. The dot1Q command associates the subinterface with the VLAN tag.

## 7. Assign IP Addresses and Default Gateways

| Device | VLAN   | IP Address    | Subnet Mask   | Default Gateway |
|--------|--------|---------------|---------------|-----------------|
| PC0    | 10 DEV | 192.168.10.10 | 255.255.255.0 | 192.168.10.1    |
| PC1    | 10 DEV | 192.168.10.11 | 255.255.255.0 | 192.168.10.1    |
| PC2    | 20 HR  | 192.168.20.10 | 255.255.255.0 | 192.168.20.1    |
| PC3    | 20 HR  | 192.168.20.11 | 255.255.255.0 | 192.168.20.1    |

Brief description: Each PC gets an address from its VLAN network and uses the router subinterface as its default gateway. In Packet Tracer: open PC → Desktop → IP Configuration.

## **9. Test Connectivity**

From PC0, open Desktop → Command Prompt and test the HR PC:

```
ping 192.168.20.10
```

From PC2, test the DEV PC:

```
ping 192.168.10.10
```

Expected result: The ping should receive replies, confirming communication between VLAN 10 and VLAN 20 through the router.

## **10. Verification Commands**

```
Switch# show vlan brief
```

```
Switch# show interfaces trunk
```

```
Router# show ip interface brief
```

Final check: VLAN 10 and VLAN 20 exist; Fa0/1–Fa0/2 are in VLAN 10; Fa0/3–Fa0/4 are in VLAN 20; Fa0/24 is a trunk; and router G0/0.10 and G0/0.20 are up/up with the configured IP addresses.

## **Conclusion**

Router-on-a-Stick allows two VLANs to communicate through one physical router interface by using separate router subinterfaces and an 802.1Q trunk.

---